

Sensationalism in Online News

Gunchang Kim*

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* Rutgers Business School, Rutgers University, Newark, NJ 07102, U.S.A. Email: gk421@scarletmail.rutgers.edu. I thank Azi Ben-Rephael, Joseph Engelberg, Byoung-Hyoun Hwang, Jin-Mo Kim, Jongho Kim, Rose Liao, Ivy Munoko for helpful comments and discussions. I thank Kenneth French for sharing U.S. research returns data, Bill McDonald for sharing Loughran and McDonald's sentiment word lists. I gratefully acknowledge the research support from the Whitcomb Center at Rutgers University. All errors are my own.

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Abstract

I find that the tone of online news is different from that of print news. In particular, the fraction of “sensational” words is significantly higher in the titles and lead paragraphs of online articles. The reason for this phenomenon potentially lies in how the success of digital media is evaluated (e.g., pageviews), which is essential for media firms in determining advertising rates. My findings suggest that media bias arises from the conflict of interests associated with different media sources, and such bias can potentially cause a price overreaction upon publication of an article and a subsequent reversal.

JEL Classifications: G14, G34, G40

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“The Web unbundles the bundle - each story becomes a separate entity that lives or dies, economically, on its own. It’s naked in the marketplace, its commercial existence meticulously measured, click by click¹.”

1. Introduction

The Web makes the underlying business model in the digital media different to that in the print media (Carr (2006)). According to Carr (2006), people purchase news stories as a package in the print media. Subscribers and advertisers pay their fees for the news package. On the other hand, in the digital media, the news package is separated into a single news story by a click. This separation evaluates the commercial existence of each news story in the market.

This difference in business models means that the criteria used to evaluate success in the digital media differ from those used in the print media. In the print media, people refer to newspaper circulation as a success evaluation criterion. On the other hand, in the digital media, success is evaluated on the basis of pageviews (the number of page requests for the website), unique visitors (the number of requests by distinct individuals for the website), and time spent by the user on a webpage (Chen, De, Hu, and Hwang (2014) and Lawrence, Ryans, and Sun (2018))².

In this paper, I investigate the question of whether the success evaluation criteria used in the digital media lead to the publication of more sensationalized stories in online editions than in print editions and how this impacts financial markets. Such bias has not been studied in the financial media literature. As the digital media has become a very popular source of information for

¹ Quote by Nicholas Carr, the former executive editor of the Harvard Business Review.

² In addition to the difference in business model between the digital and print media, newspaper circulation is hard to use as a success evaluation criterion in the digital media because it is challenging to measure digital media circulation (Barthel (2019)). According to Barthel (2019), in recent years, *The Wall Street Journal*, *The New York Times*, and *The Washington Post* have not fully reported digital media circulation to the Alliance for Audited Media, which audits the circulation of many large North American newspapers.

investors, this unexplored but important question can provide valuable information for them.

Media firms have strong economic incentives to increase pageviews and unique visitor numbers for their online editions because both are vital in determining advertising rates³ (Evans (2009) and Chakraborty, Paranjape, Kakarla, and Ganguly (2016)). For example, *The Los Angeles Times* editors examine the previous day's pageviews during daily meetings of the News section⁴. Pageviews and unique visitors are also essential criteria for journalists because they can serve as either a motivation for higher compensation, a threat to maintaining their occupation, or both⁵. However, such incentives are relatively weak in the print media.

These economic incentives caused by the conflict of interests associated with different media sources raise the empirical question of whether the *tone* of online news is different from that of print news. To examine the question mentioned earlier, I obtain English-language news articles issued in online and print editions by four influential national newspapers—*The Wall Street Journal* (WSJ), *The New York Times* (NYT), *USA Today*, and *The Washington Post* (WSP)—from 2007 to 2018.

By using Loughran and McDonald's sentiment word lists⁶, I identify the negative and

³ Besides the online editions of newspapers, advertisers can choose the cost per click (CPC) or cost per view (CPV) model to advertise on other online platforms, such as YouTube. Similarly, the cost per impression (CPI) and the click through rate (CTR) models are used to set advertising revenue.

⁴ Quote by Jimmy Orr, the former *LA Times* digital managing editor (Fischer (2014)). Similarly, *The Wall Street Journal.com* (*The WSJ.com*) states that they reached 42 million digital readers during the first quarter of 2017 through *The WSJ.com* Audience Profile.

⁵ Authors in Seeking Alpha, an online news provider about financial markets, receive \$10 per 1000 pageviews as compensation (Chen, Du, Hu, and Hwang (2014)). An Australian journalist was fired from the firm because his reviews did not make sufficient hits on the websites (Moses (2007)). In addition, with recent developments in technology, it has become easier to ascertain the number of article views (e.g., A journalist at *Philadelphia Inquirer* can easily obtain information about the number of clicks an article achieves sorted by journalist names (Anderson (2011))).

⁶ Loughran and McDonald's sentiment word lists (Loughran and McDonald (2011)) are very widely used to measure news sentiment in the financial media literature (e.g., Gurun and Butler (2012), Ahern and Sosyura (2014), Chen, De, Hu, and Hwang (2014), Solomon, Soltes, and Sosyura (2014), and Bradshaw, Wang, and Zhou (2015)).

positive words in each article. I define *tone* as the number of positive words minus the number of negative words over the total number of words. I find that there is a significant difference in *tone* between the articles published in the online editions and those published in the print editions. This difference could potentially arise as a result of a strategic change in the use of sensational words in online editions, as compared to print editions, in order to increase article views⁷.

To examine the possibility of a strategic change in the use of sensational words aimed at inducing readers to click online articles, I use the characteristics of online news editing⁸. One of the most significant differences between online edition articles and their print edition counterparts is the way they are edited (Kim (2011)). According to Kim (2011), in online publications, readers can only see the title and/or lead paragraph when deciding which articles to read. On the other hand, in print editions, readers often have a chance to see the entire article. Hence, there is a greater emphasis on editing the titles and lead paragraphs of online edition articles. My results show that a significantly higher fraction of both negative and positive words is found in the titles and lead paragraphs of online edition articles, which suggests evidence of media firms seeking to maximize the number of article views.

Greater use of negative and positive words makes the tone more volatile in online editions than in print editions. By using share turnover as a proxy for overall investors' attention, I find that the tone in online editions is more volatile for firms with a higher share turnover. I also use extreme abnormal return and media coverage as two other proxies for investor attention and obtain similar

⁷ However, the difference in *tone* can also arise from the different articles published in the online and print editions. I mitigate the concern by investigating same-title articles published in both online and print editions by the same journalist(s) in the later parts of the paper.

⁸ As pageview data are mostly proprietary data, I use the characteristics of online news editing to infer the intention to induce readers to click online articles by increasing the use of sensational words.

results. These results suggest evidence of newspapers trying to capture greater viewer attention by using catchier words in their online edition articles for firms with greater investor attention. Media firms and journalists could potentially increase article views by attracting readers with catchier words.

In capital markets, investors do not weigh all available information equally. According to the limited attention theory, more salient news produces a greater immediate market response (Hirshleifer and Teoh (2003) and Hirshleifer, Lim, and Teoh (2011))⁹. I find that online news tone has a more significant impact on market response than print news tone. The difference between online and print news *tone* is statistically significant and economically meaningful. A one standard deviation increase in online edition *tone* has an approximately eight times greater impact on market response than a corresponding increase in print edition *tone*.

Evidence from the psychology literature shows that the first impression from a news headline creates a lasting effect that is difficult to overcome, even after conflicting information is provided by the text following the headline (Rabin and Schrag (1999), Konnikova (2014), and Ecker, Lewandowsky, Chang, and Pillai (2014)). On the basis of psychological theory and evidence, Huang, Nekrasov, and Teoh (2018) find that in relation to earnings news, higher “headline salience” has a more significant impact on market response than lower headline salience. In support of these results, recent accounting studies show that the location of the information provided matters in terms of market response¹⁰. Consistent with the results from prior studies, I

⁹ Empirical evidence is documented in Cohen, Diether, and Malloy (2013) and Elliott, Rennekamp, and White (2015).

¹⁰ Files, Swanson, and Tse (2009) find that market reaction after restatement is strongest when the information is provided in the headline than when the information is provided in the body of the text or in footnotes. Bowen, Davis, and Matsumoto (2005) show similar results by comparing the location provided in pro forma earnings versus GAAP earnings.

find that the *tone* of the title and lead paragraph of both the online and print versions of an article has a greater impact on market response than the *tone* of both versions of the whole article. This effect is more pronounced in online edition articles.

In recent years, the quantity of print media coverage has declined substantially following the emergence of digital media¹¹. To mitigate the concern that the higher number of different articles issued online version causes the difference in *tone*, I examine articles published in online and print versions with the same title and written by the same journalist(s). My results remain robust under the subsample. Surprisingly, there is not only a higher use of negative and positive words in the lead paragraphs of the online editions but also a clear switch to using sensational words for the text following the lead paragraphs. I do not find evidence of a significantly higher use of salient words for the text following the lead paragraphs in online editions as compared to print editions. These results also support the evidence showing a higher use of catchier words in the titles and lead paragraphs of online versions as compared to print versions in order to increase article views.

A more salient news tone in the title and lead paragraph of the online version of an article as compared to the print version can potentially cause a price overreaction upon publication of the article and a subsequent reversal. In line with Gurun and Butler (2012) and Golez and Karapandza (2019), I create a daily long-short portfolio on the basis of the *abnormal online tone*, which is the difference between the online tone and print tone of the titles and lead paragraphs. Then, I long

¹¹ According to the Alliance for Audited Media report, print news circulation decreased by 39% from 1998 to 2013 for the 15 largest newspapers in the U.S. Some international newspapers also decided to stop issuing a print edition and only issue an online edition (e.g., *Seattle Post-Intelligencer* (U.S.), *La Presse* (Canada), *The Independent* (U.K.), and *Taloussanomati* (Finland)). According to Abernathy (2018), print readers are vanishing at a faster rate than print newspapers, and this phenomenon has appeared to be accelerating in recent years.

stocks with a relatively low abnormal online tone and short stocks with a relatively high abnormal online tone. To see a clearer effect, I construct a portfolio using the stocks related to the articles that are published on the same date in online and print versions with the same titles and written by the same journalist(s). This long-short portfolio generates an abnormal return of approximately 1% per month after controlling for well-known risk factors¹².

I further examine how favorable and unfavorable news about firms affects the tone of the titles and lead paragraphs in the online editions as compared to the print editions. I examine the tone of the post-earnings announcement using the articles issued within one day of the announcements. I define *forecast error* as the value of actual earnings minus the mean value of the analyst forecasts. I partition entire announcements into three categories on the basis of forecast error: Positive announcements, neutral announcements, and missed announcements are when the *forecast error* is greater or equal to five cents, greater or equal to zero cents and less than five cents, and less than zero cents, respectively.

My results show that the *tone* of the titles and lead paragraphs in the online versions is significantly higher (lower) than that in the print versions for positive (missed) announcements. These results suggest that online editions tend to overstate their news stories more than print editions (i.e., online editions report news that is favorable (unfavorable) to a firm more favorably (unfavorably) than print editions). Compared with print versions, there is a significantly higher fraction of negative and positive words used in the titles and lead paragraphs of online versions after neutral announcements. These results further support the argument that there is a dramatic

¹² 1% excess return per month is comparable to the amount obtained in other studies from a long-short portfolio construction based on media bias (e.g., Gurun and Butler (2012) obtained an excess return of 5.52% per year, Ahern and Sosyura (2015) achieved 1% excess return per month, and Golez and Karapandza (2019) obtained 1.4% excess return per month).

change in the use of catchy words in the online versions as compared to the print versions.

I believe that my paper contributes to the growing financial media literature¹³. To the best of my knowledge, my study is the first to shed light on the pageview journalism issue of online news in the financial media literature. Recent studies show that media slant often varies due to conflicts of interest¹⁴. My findings support previous evidence indicating that news characteristics change due to the conflict of interests associated with different media sources. The unique contribution of my study is that it raises the question of regulatory issues in the media industry and whether pageview journalism is a proper way to evaluate the success of digital media¹⁵.

Many recent studies on financial media do not stipulate whether their news data are sourced from the online edition, the print edition, or both. Some studies only use print editions to answer research questions. My findings suggest that researchers should use online editions for future studies as the tone of online news has a greater impact on financial markets than the tone of print news.

Additionally, I believe that my findings have new implications regarding the information provided via the Web beyond the financial media literature. Many recent studies show that information shared or disseminated through the Internet (e.g., social media postings and product

¹³ Recent studies show that business news plays an essential role in disseminating information and has a significant impact on financial markets (Tetlock (2007), Tetlock, Saar-Tsechansky and Macskassy (2008), Fang and Peress (2009), and Engelberg and Parsons (2011)).

¹⁴ Gurun and Butler (2012) show that firms' local advertising expenditures result in a favorable slant in local media stories on local firms (see also Dyck and Zingales (2003) and Gentzkow and Shapiro (2010) for the effect of advertising on news slant). DellaVinga and Hermle (2017) address how the media firm's ownership affects favorable movie ratings. You, Zhang, and Zhang (2017) provide evidence of the government's impact on market-oriented media and show how such an impact affects market prices. Xu (2019) finds that journalist connection leads to the issuance of more favorable news. Golez and Karapandza (2019) address home county media slant and equity prices.

¹⁵ As more sensationalized online news stories lead to a price overreaction upon publication of articles, there should be specific guidelines to follow for online edition articles: for example, restrict access to the number of article views to the journalist(s) or introduce a content rating system that allows readers to evaluate various components of an article, including the degree of sensationalism and credibility.

reviews) has a significant impact on stock markets¹⁶. However, it is not hard to find sensationalism in the information shared via the Web. Sensationalism on the Internet can often become a source of fake news and fake reviews, and it could become a tool for firms to share negative stories about their industry peers on social media to capture investors' attention¹⁷. Investors can make use of my findings to make wise judgments when using information shared through the Web.

The rest of this paper is organized as follows: The next section describes the data, section 3 discusses the empirical results, and section 4 concludes the paper.

2. Data

I download English-language online and print edition news articles for S&P 500 firms from Factiva. The sources of these news articles are four national newspapers whose online and print editions are available on Factiva: *The Wall Street Journal (WSJ)*, *The New York Times (NYT)*, *USA Today*, and *The Washington Post (WSP)*¹⁸. Table 1 provides a list of the online and print editions of newspapers used in the study¹⁹.

The sample period is from 2007 to 2018. I start collecting news data from 2007 because it was in that year that Factiva started providing full coverage of articles published in both the online

¹⁶ See Blankespoor, Miller, and White (2014), Chen, De, Hu, and Hwang (2014), Green, Huang, Wen, and Zhou (2019), and Huang (2019), among others.

¹⁷ Kogan, Moskowitz, and Niessner (2019) find that fake news is more sensational than non-fake news. Filieri (2016) finds that reviews containing sensational titles are more likely to be recognized as untrustworthy by customers. Cao, Fang, and Lei (2019) provide evidence that firms make use of their Twitter account to post negative stories in order to compete with their rivals in their product market.

¹⁸ Solomon, Soltes, and Sosyura (2014) use the same four national newspapers to examine how media coverage of mutual holdings affects investors' money allocation across funds. Fang and Peress (2009) also focus on the same newspapers to discuss media coverage and a cross-section of stock returns.

¹⁹ In addition to *The Wall Street Journal Online*, WSJ also launched the WSJ Pro edition. However, I do not include it in the study because it offers specialized content aimed at professionals in specialized industries through the online platform (Pompeo (2015)).

and print editions of the four newspapers included in the study²⁰. Following Engelberg (2008) and Ahern and Sosyura (2014), I use the Intelligent Indexing code assigned by Factiva to download news articles so that I can obtain news articles sufficiently relevant to my sample firms²¹. I separate each article into the title, the lead paragraph, and the text following the lead paragraph using Factiva's field tags.

To measure news sentiment, I obtain Loughran and McDonald's sentiment word lists from their websites²². Loughran and McDonald's sentiment word lists are specially designed to measure slant in the financial context. I classify negative and positive words and define a few news sentiment variables as follows: *Tone* is the number of positive words minus the number of negative words over the total number of words; *Frac neg* is the number of negative words over the total number of words; *Frac pos* is the number of positive words over the total number of words.

$$Tone = \frac{\# of pos - \# of neg}{\# of words} \times 100, \quad Frac\ neg\ (Frac\ pos) = \frac{\# of neg\ (pos)}{\# of words} \times 100$$

As the editing of titles and lead paragraphs in the online editions is essential to compete for readership, I define news sentiment variables focusing on the whole article, the title and lead paragraph, and the text following the lead paragraph separately. I further define *Coverage* as the number of articles issued by the newspaper in a quarter.

Panel A of Table 2 reports the summary statistics for news characteristics in the online and

²⁰ The quarter when Factiva started providing full coverage of articles published in both online and print editions is different for the four newspapers (WSJ from Q3 2008, NYT from Q1 2012, The USA Today from Q3 2016, and WSP from Q4 2007). To compare the news characteristics of the online editions to those of the print editions, I only include quarters for which I can obtain both online and print editions at the newspaper-quarter level.

²¹ Using Factiva's Intelligent Indexing code to search articles provides more relevant articles than a less precise keyword search (Ahern and Sosyura (2014)).

²² Available at <https://sraf.nd.edu/textual-analysis/resources/>

print editions. I find that the tone of articles issued in the online editions is significantly different to the tone of articles issued in the print editions. The average *tone* of the articles published in the online editions is -1.091. On the other hand, the average tone of the print edition articles on the same firms over the same sample period is -1.172, suggesting that compared with the print editions, the online editions can potentially strategically change the use of salient words to compete for readership.

In addition to the tone difference in the articles as a whole, the titles and lead paragraphs of the articles issued in the online editions have a significantly higher *Frac neg* and *Frac pos* than articles issued in the print editions. The higher use of salient words in the titles and lead paragraphs of online editions can potentially arise from the newspapers' desire to increase article views. In terms of coverage, Panel A of Table 2 also shows that on average, three more articles about my sample firms are published per quarter in the online editions than in the print editions.

In addition to the news data, I obtain financial statements, stock returns, and institutional ownership data from the CRSP/Compustat Merged (CCM), CRSP, and Thomson-Reuters Institutional Holdings (13F) databases, respectively. Panel B of Table 2 reports the summary statistics for the firm-level variables. The appendix provides detailed definitions of the variables. The mean (median) value of the abnormal stock return is 0.382 (0.295), indicating that the sample firms had a boom period over the sample period.

3. Empirical Results

3.1. Relative Tone of Online News

In this section, I present the empirical model for estimating the relative tone of the online

editions compared to the print editions and the results obtained from that model.

$$News\ characteristics_{i,j,t} = \alpha + \beta_1 Online_{j,t} + \beta_2 X_{i,t} + \gamma_i + \varphi_t + \omega_j + \varepsilon_{i,j,t} \quad (1)$$

where *News characteristics*_{*i,j,t*} denotes the news sentiment variables (*Tone*, *Frac neg*, and *Frac pos*) for firm *i* by newspaper *j* at time *t*. *Online*_{*j,t*} is an indicator variable that takes the value of one if a news article is published in the online edition of newspaper *j*. The coefficient of *Online* captures the relative news characteristics of online editions compared to print editions. *X*_{*i,t*} includes various firm characteristic variables that could potentially affect news characteristics for firm *i*, such as firm size (log(*assets*)), growth opportunities (Sales growth rate and Tobin's *q*), capital expenditure (CAPEX / asset), profitability (ROE), financial leverage (Leverage), stock return (Abnormal stock return), and institutional ownership (Institutional ownership) (Gurun and Butler (2012), You, Zhang, and Zhang (2017), and Kedia and Kim (2020)).

I control for the information environment of the firm by including *log(assets)* and *Institutional ownership*. The information environment of the firm could potentially affect news characteristics as media reports could play an external governance role (Gurun and Butler (2012)). I also control for several accounting variables (*Sales growth rate*, *Tobin's q*, *CAPEX / asset*, *ROE*, and *Leverage*) that capture the financial condition of a firm. If the financial condition of the firm is good, it is more likely to be covered with a favorable slant by the media. Additionally, I control for stock return variables (*Abnormal stock return*). Stock return incorporates news sentiment (Tetlock (2007)). An increased stock return could affect the firm-specific investor sentiment.

I include firm fixed effects (γ_i), quarter fixed effects (φ_t), and newspaper fixed effects (ω_j) to control for unobservable time-invariant firm characteristics, time trends, and journalistic differences across the newspapers. I compute robust standard errors adjusted for heteroscedasticity

clustered by the firm level (Engelberg and Parsons (2011) and Gurun and Butler (2012)).

Panel A of Table 3 presents the estimates of ordinary least squares (OLS) regressions of article characteristics on the indicator for online article. In model (1), I find that the coefficient on *Online* is positive and significant. The coefficient of 0.061 on *Online* implies that online editions publish articles that are on average 4.63%²³ more favorable than those published in print editions. This result supports the evidence indicating that the tone of online news is significantly different from that of print news. In support of the result in model (1), I find a negative (positive) and significant coefficient on *Online* in model (4) (model (7)), suggesting that the use of negative and positive words in online editions is significantly different from that in print editions.

However, these results raise the question of whether online editions always issue more favorable articles than print editions or whether the tone of online news changes more dynamically than the tone of print news. To answer the question, I partition my sample into an extremely high group and an extremely low group by using the value of the news sentiment variables. Articles belong to the extremely high (low) tone group if the news article tone falls into the top (bottom) 20% tone by each year and newspaper. Similarly, the articles are classified into an extremely high (low) *frac neg* group and an extremely high (low) *frac pos* group²⁴.

In model (2), I find a positive and significant coefficient on *Online* for the extremely high tone group. The coefficient of 0.015 on *Online* for the extremely high tone group indicates that on average, online editions issue 1.61% more favorable articles than print editions. On the other hand,

²³ The calculation is based on the coefficient estimate of *Online* (0.061) over the unconditional mean value of *Tone* (-1.318).

²⁴ An article belongs to the extremely high (low) *frac neg* group if the fraction of negative words in the article falls into the top (bottom) 20% *Frac neg* by each year and newspaper. An article belongs to the extremely high (low) *frac pos* group if the fraction of positive words in the article falls into the top (bottom) 20% *Frac pos* by each year and newspaper.

in model (3), I find a negative and significant coefficient on *Online* for the extremely low tone group. On average, the articles published in online editions are 0.88% more unfavorable than the articles published in print editions in the extremely low tone group. These results suggest that online news tone changes more dynamically than print news tone.

As the editing of titles and lead paragraphs in the online editions is critical in the competition for readership, I additionally define news sentiment variables focusing on the titles and lead paragraphs. Panel B of Table 3 presents the results of OLS regressions to estimate the characteristics of the titles and lead paragraphs on the indicator for online article. In model (7), I find a positive and significant coefficient of 0.044 on *Online*, which suggests that to increase article views, online edition articles, on average, use 4.28% more positive words in their titles and lead paragraphs than their corresponding print versions.

The positively significant coefficient on *Online* in model (5) and the negatively significant coefficient on *Online* in model (6) suggest that online edition articles contain 5.07% more negative words in their titles and lead paragraphs in the extremely high frac neg group and 11.20% fewer negative words in their titles and lead paragraphs in the extremely low frac neg group. I find a positive and significant coefficient on *Online* in model (8) and a negative and significant coefficient on *Online* in model (9). These findings suggest that online edition articles, on average, contain 3.10% more positive words in their titles and lead paragraphs in the extremely high frac pos group and 10.37% fewer positive words in their titles and lead paragraphs in the extremely low frac pos group.

Similar to the results presented in Panel A of Table 3, the use of sensational words in the titles and lead paragraphs changes more dynamically in the online edition articles than in the print

edition articles.

3.2. Tone Volatility and Investor Attention

In this section, I first examine the tone volatility of online edition articles relative to print edition articles. To do this, I define *Tone volatility* as the standard deviation in the tone of the articles issued by a newspaper in a quarter. *Frac neg volatility* is the standard deviation in the *frac neg* of articles issued by the newspaper in a quarter. *Frac pos volatility* is the standard deviation in the *frac pos* of articles issued by the newspaper in a quarter.

Panel A of Table 4 presents the estimates of OLS regressions of article characteristics on the indicator for online article. In model (1), I find a positive and significant coefficient on *Online*, providing evidence that, on average, the tone of the articles published in the online editions is around 3.28% more volatile than that of articles published in print editions. In support of the result in model (1), I find a positive and significant coefficient on *Online* in models (2) and (3), suggesting that the use of negative and positive words changes more dynamically in online edition articles than in print edition articles.

I further investigate whether higher tone volatility is associated with greater investor attention. The results presented in Panel A illustrate that the change in tone is more volatile in online news than in print news. If the higher tone volatility aims to induce readers to click the online articles, then it is likely to be more pronounced for firms with greater investor attention.

I use share turnover (Hou, Peng, and Xiong (2006)) as a proxy for investor attention. Trading activity is highly associated with the active attention paid to a firm by investors (Loh (2010)). I divide the sample by the quartile values of share turnover. To see a clearer effect, I classify firms

into a high (low) group if the value of their share turnover falls into the top (bottom) quartile²⁵. The regression results are reported in Panel B1 of Table 4. Consistent with my expectations, models (1) and (2) show that the estimate of *Online* is 0.077 (t-stat = 4.12) for the firms with higher share turnover and 0.025 (t-stat = 1.34) for the firms with lower share turnover. This result supports the idea that the tone of online editions is more volatile than that of print editions for firms with greater investor attention. I find similar results using *frac neg volatility* and *frac pos volatility* in models (1) and (2) in Panels B2 and B3.

To provide additional evidence, I use extreme return (Barber and Odean (2007)) and media coverage (Liu, Sherman, and Zhang (2014) and Drake, Jennings, Roulstone, and Thornock (2016)) as proxies for investor attention. To examine whether higher tone volatility is associated with greater investor attention, I partition the sample by the quartile values of extreme return and media coverage.

Models (3) and (4) indicate that the estimates on *Online* are positive and significant, and the effect is substantial for firms with high extreme return. Similarly, I find a stronger effect on the firms with high media coverage in models (5) and (6). Overall, these results suggest that the higher tone volatility in online editions is associated with firms that attract greater investor attention and could potentially be aimed at inducing readers to click articles.

3.3 Market Response to Online and Print News

Next, I examine whether online news tone has a more significant impact on market response than print news tone. My dependent variable is two-day cumulative abnormal returns

²⁵ To see a clearer effect, Tetlock, Saar-Tsechansky, and Macskassy (2008) also label positive (negative) news using the bottom (top) quartile of the standardized number of negative words in news articles.

around article issuance date (CAR [0,1]). Following You, Zhang, and Zhang (2017), I use the daily average value of the news sentiment variables as my explanatory variables. I define *Online tone* (*Print tone*) as the average daily tone of the articles issued in the online (print) editions. *Online frac neg* (*Print frac neg*) is the average daily fraction of negative words in the online (print) editions. *Online frac pos* (*Print frac pos*) is the average daily fraction of positive words in the online (print) editions. I define these variables in the whole article and in the title and lead paragraph separately. I also include the same control variables as those employed in Table 3. I include firm fixed effects and quarter fixed effects to control for unobservable time-invariant firm characteristics and time trends and cluster robust standard errors by firm level.

Table 5 presents the estimates of OLS regressions of CAR [0,1] around article issuance date on online article tone and print article tone. In model (1), I find a positive and significant (insignificant) coefficient on *Online tone* (*Print tone*), which suggests that online news tone has a greater impact on market response than print news. The estimate of *Online tone* is around seven times larger than that of *Print tone*. The difference in coefficients estimates between the two is statistically significant and economically meaningful. These coefficients imply that a one standard deviation increase in *Online tone* and *Print tone* (0.017 and 0.015, respectively) is associated with an increase in CAR [0, 1] of 0.099% and 0.012%, respectively. In models (2) and (3), I find similar results, namely that the market responds more negatively to *Online frac neg* than to *Print frac neg* and more positively to *Online frac pos* than to *Print frac pos*.

On the basis of psychological evidence, salience in the titles and lead paragraphs of articles can potentially have a greater impact on market response. In models (4) to (6), I perform regressions using the tone of the titles and lead paragraphs and find similar results. In model (4),

the estimate of *Online tone* is 0.044 and that of *Print tone* is 0.006, which implies that a one standard deviation increase in the *Online tone* and *Print tone* of the titles and lead paragraphs (0.029 and 0.027, respectively) results in an increase in CAR [0,1] of 0.128% and 0.016%, respectively. Consistent with evidence from prior studies, these increases in CAR [0,1] due to the tone of the title and lead paragraph are higher than the increases in CAR [0,1] due to the tone of the whole article in both online and print editions. This effect is more pronounced in the online edition articles. These results provide evidence of the importance of the titles and lead paragraphs, especially in the online editions.

However, as online news publication is published on a real-time basis, the question remains as to whether, generally, the issuance of online news prior the issuance of print news causes a difference in market reactions²⁶. To mitigate the concern that the publication of online news prior to the publication of print news makes a difference in terms of market response, in section 3.5, I examine the same-title articles by the same journalist(s) issued on the same date in both the online and print editions. The results provided in section 3.5 show evidence that the tone difference between online and print news causes a price overreaction upon publication of an article and a subsequent reversal.

Overall, these results suggest that online news tone has a more significant impact on market response than print news²⁷.

²⁶ One of the difficulties I face in the study is that for most of my sample, I know the date an article was published but not the time. Hence, I focus on the subsample comprised of same-title articles by the same journalist(s) issued on the same date in online and print editions. This subsample can mitigate the timing issue because each pair of articles in the subsample was published on the same date in print and online editions; as print news is issued early in the morning, there is less concern that the online edition article was issued earlier than the print edition article.

²⁷ Gurun and Butler (2012) control for media coverage as it can potentially have a higher impact on investor's recognition. Controlling for online and print edition coverage does not change my results qualitatively.

3.4. Relative Coverage of Online News and Same Title Articles by The Same Journalist(s)

In this section, I address the concern that despite the evidence that the tone of online news is significantly different from that of print news, this could be due to the higher number of articles issued in online editions compared with print editions. First, I examine the relative coverage of the online editions as compared to that of the print editions. Table 6 presents the estimates of OLS regressions of *Coverage* on the indicator for online article. In model (1), I find a positive and significant coefficient on *Online*, providing evidence that, on average, online editions issue around 3.68 more articles per quarter than print editions. The result in Table 6 is generally consistent with Fang and Peress (2009). Media coverage is overwhelmingly affected by firm size (i.e., larger firms have a greater chance of being featured in the media). The abnormal stock return does not have a significant impact on the likelihood of media coverage.

To mitigate the concern that tone differences arise from the higher number of different articles published in online editions as compared to print editions, I search for same-title articles by the same journalist(s)²⁸. Panel A of Table 7 presents the summary statistics for the news characteristics of 23,762 same-title articles by the same journalist(s) published in both online and print editions²⁹. I find that the tones of the lead paragraphs for the same-title articles by the same journalist(s) are significantly different in the online and print editions. The average *tone* of the lead paragraphs published in the online editions is -1.588; on the other hand, the average tone of the

²⁸ Not all articles provided by Factiva have the name(s) of the journalist(s) on them. When searching the same-title articles by the same journalist(s), I exclude articles if I cannot see the name(s) of the journalist(s) on them (e.g., articles with the name(s) of the journalist(s) missing or articles issued by the editorial board or Associated Press). If an article is written by multiple journalists, I include the article only if all the authors are the same in both the online and print editions. Also, I exclude articles with commonly used titles (e.g., Business Watch and Business Digest) because it is hard to differentiate whether they are based on the same event. These strict conditions give more insights into the characteristics of news articles published in the online editions relative to those published in the print editions.

²⁹ Around 80% of the sample firms are covered in 23,762 same-title articles by the same journalist(s).

lead paragraphs issued in the print editions for the same-title articles by the same journalist(s) over the same sample period is -1.485. This result provides evidence that my results are not driven by a higher number of online edition articles or by different events covered in online and print editions.

In addition, the *Frac neg* of the lead paragraphs is significantly higher in the online editions than in the print editions. The average *Frac neg* of the lead paragraphs published in the online editions is 2.466. In contrast, the average *Frac neg* of the lead paragraphs issued in the print editions for the same-title articles by the same journalist(s) over the same sample period is 2.345. This result suggests evidence of increasing article views by using more sensational words in the online editions.

It is noticeable that there is no significant difference between the online and print editions in terms of the news characteristics of the text following the lead paragraph. This result supports the evidence showing that there is a strategic change in the use of salient words, especially in the lead paragraphs of same-title articles by the same journalist(s), so that media firms and journalist(s) can potentially increase article views.

Panel B of Table 7 reports the estimates of OLS regressions of the article characteristics of same-title articles by the same journalist(s). I find a negatively significant coefficient of -0.007 on *Online* in model (1). This coefficient indicates that in regard to same-title articles by the same journalist(s), 0.54% more unfavorable articles are issued in the online editions than in the print editions. The effect is more pronounced in the titles and the lead paragraphs. The coefficient of -0.065 on *Online* in model (4) indicates that 3.95% more unfavorable articles are published in online editions than in print editions. These results suggest that the tone of both the whole articles and the

titles and lead paragraphs are significantly different in the online and print editions even when the articles are written by the same journalist(s) and have the same title.

In models (5) and (6), I find a positive and significant coefficient on *Online*. The coefficients in models (5) and (6) suggest that compared with the print editions, the online editions contain 3.30% more negative words and 2.00% more positive words in their titles and lead paragraphs. These results provide additional evidence of the higher use of both negative and positive words in online edition articles as compared to print editions, which can potentially be attributed to the digital media evaluation criteria. Overall, the results in this section provide evidence of a strategic change in the use of striking words.

3.5. Online News Trading Profits

Next, I examine online news trading profits. The results so far suggest that more salient words are used in the titles and lead paragraphs of online edition articles than in the titles and lead paragraphs of print edition articles. Such salience can potentially cause a stock price overreaction at the time an article is issued. To examine this possibility, in line with Gurun and Butler (2012) and Golez and Karapandza (2019), I define abnormal online tone as the difference between the online tone and print tone of the titles and lead paragraphs. I construct a daily long-short portfolio using the stocks related to the articles that are published on the same date online and in print with the same title and by the same journalist(s)³⁰. I long stocks with a low abnormal online tone and short stocks with a high abnormal online tone. I hold both long and short positions for one day after the investment, and the portfolio is rebalanced daily.

³⁰ Among the same-title articles by the same journalist(s) published in both online and print editions, around 71.6% are issued on the same date. For portfolio construction, I focus on the articles issued on the same date to mitigate the concern that different publication dates in online and print editions could potentially affect the results.

Panel A of Table 8 presents the daily risk-adjusted returns from a daily long-short portfolio. The investment is made immediately after portfolio construction without waiting. I examine three different factor models using the market model, the Fama and French (1993) three-factor model, and Carhart's (1997) four-factor model. In models (1) to (3), I find a positive and significant alpha in equally weighted portfolios, suggesting that the average excess return is around 1% per month after controlling for well-known risk factors. The excess return I obtain is similar in magnitude to that obtained by Ahern and Sosyura (2015) and Golez and Karapandza (2019) from their long-short portfolio construction based on media bias. However, it is greater than the excess return of 5.52% per year in Gurun and Butler (2012). The low value of the R-squared statistics suggests that the risk of trading strategy is firm specific because I construct a portfolio on the basis of firm-specific news articles (Tetlock, Saar-Tsechansky, and Macskassy (2008)). I find similar results in models (4) to (6), providing evidence that my results do not change qualitatively in value-weighted portfolios³¹.

There is the possibility that other news articles, apart from the same-title articles, about the same firm issued on the same date with the same titles could potentially affect the results provided in Panel A of Table 8. To mitigate this concern, I label days among the sample used in Panel A as low media coverage days if, excluding the same-title articles, no more than one article was issued in either print or online editions. In an untabulated analysis, I find that my results do not change qualitatively in low daily media coverage days. This finding suggests that the results provided in Panel A are less likely to be driven by other news articles about the same firm issued on the same

³¹ My sample period includes the financial crisis period. The results provided in Panel A of Table 8 do not change qualitatively after excluding the articles issued in the financial crisis period. The financial crisis period is defined on the basis of the definition in U.S. Business Cycle Expansions and Contractions from the National Bureau of Economic Research (NBER). Available at <https://www.nber.org/cycles.html>

date with the same title.

Panel B of Table 8 presents the excess return for the daily risk-adjusted returns from a daily long-short portfolio. The investment is made after portfolio construction following a waiting period, which can provide an insight into how long the reversal continues. In models (4) to (6), both the magnitude of excess return and the significance of excess return decrease in the value-weighted portfolios if the investment is made following a one-day wait after portfolio construction. In cases where the investment was made following a two or three day wait after portfolio construction, I do not obtain a significant excess return. These results suggest that there is an immediate price overreaction upon publication of an article and a subsequent short-term reversal. In models (1) to (3) using equally weighted portfolios, the reversal continues one more day but provides similar results to the value-weighted portfolios.

Overall, the results suggest that sensationalism in online editions can potentially cause a price overreaction upon publication of an article and a subsequent reversal, and the online edition article is less likely to provide more value relevant information than the print edition article.

3.6. Post Earnings Announcement Tone

Finally, I investigate how favorable and unfavorable news to firms affects the relative use of negative and positive words in the titles and lead paragraphs of online edition articles as compared to print edition articles. If the online editions strategically change the words in the titles and lead paragraphs of articles to compete for readership, I expect a relatively higher use of positive (negative) words in the online editions when news that is favorable (unfavorable) to a firm comes out.

To examine this, I obtain quarterly earnings and analysts forecasts for the sample firms from the I/B/E/S database. Following Kasznik and McNichols (2002) and McVay, Nagar, and Tang (2006), I define forecast error as the value of actual earnings minus the mean value of the analyst forecasts. I partition all announcements into three categories on the basis of forecast error: Positive announcements are when the forecast error is greater or equal to five cents, negative announcements are when the forecast error is greater or equal to zero cents and less than five cents, and missed announcements are when the forecast error is and less than zero cents. I focus on the news articles issued within one day of the earnings announcements in the online and print editions.

Table 9 reports the estimates of OLS regressions of the article characteristics on the indicator for online article. In model (1), I find a positive and significant coefficient on *Online*. On the other hand, in model (7), I find a negative and significant coefficient on *Online*. These results provide evidence that the tone of the titles and lead paragraphs in the online editions is, on average, 13.36% more favorable (7.74% less favorable) than that in the print editions after positive (missed) announcements. These results suggest that online editions issue articles are more favorable (unfavorable) in tone than articles in print editions when news that is favorable (unfavorable) to a firm comes out.

It is noticeable to see the results for neutral announcements. In models (5) and (6), I find a positive and significant coefficient on *Online*. These results indicate that, on average, online edition articles contain 4.94% more negative words and 10.36% more positive words in their titles and lead paragraphs than print edition articles even after neutral announcements. Overall, the results provided in this section provide additional evidence suggesting that to increase article views, newspapers use more sensational words in the titles and lead paragraphs of their online

editions than in the titles and lead paragraphs of their print editions³².

4. Conclusion

This paper examines sensationalism in online news. I find that online edition articles use more sensational words than print edition articles, especially in their titles and lead paragraphs, suggesting that newspapers use sensational words to increase online article views. This result is robust under the subsample of articles published in online and print editions with the same title by the same journalist(s). My findings highlight that news characteristics change due to conflicts of interest that arise as a result of the criteria used to evaluate digital media. The distinctive contribution of my study is that my findings highlight regulatory issues that raise the question of whether measuring the number of article clicks is the best way to evaluate the success of digital media.

I provide new evidence that tone difference in online and print articles is statistically significant and economically meaningful. The excess return I obtain from a long-short portfolio shows that a more salient tone in online news as compared to print news can potentially cause a price overreaction upon publication of an article and a subsequent reversal. These findings are important because digital media has become a very popular source of information for investors. Investors can make use of my findings to discount sensational online news. My results additionally suggest that researchers should differentiate online news from print news in future studies as the former has a greater impact on financial markets than the latter.

³² My results do not change qualitatively after excluding the financial crisis period from the sample. The financial crisis period is defined on the basis of the definition in U.S. Business Cycle Expansions and Contractions from the National Bureau of Economic Research (NBER). Available at <https://www.nber.org/cycles.html>

I believe that my findings have considerable implications for the information provided through the Web beyond the online news. Sensationalism in the information shared via the Web, such as social media postings or product market reviews, often becomes a source of fake news or fake reviews and could have a significant impact on stock markets. Investors can take advantage of my findings to make decisions judiciously.

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Table 1
List of Newspapers in Online and Print Editions

This table presents the list of the newspapers and their online and print editions used in the study.

Newspaper name	Online edition	Print edition
The Wall Street Journal	The Wall Street Journal Online	The Wall Street Journal
The New York Times	NYTimes.com	The New York Times
USA Today	USA Today Online	USA Today
The Washington Post	Washington Post.com	The Washington Post

Table 2
Summary Statistics

This table presents the summary statistics for a sample of S&P 500 firms from 2007 to 2018. The sample includes news articles issued in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Tone* is the number of positive words minus the number of negative words over the total number of words in the whole article (or title and lead paragraph) as a percentage. *Frac neg* is the number of negative words over the total number of words in the article (or title and lead paragraph) as a percentage. *Frac pos* is the number of positive words over the total number of words in the article (or title and lead paragraph) as a percentage. The title and lead paragraph are classified on the basis of Factiva field tags. *Coverage* is the number of articles issued by the newspaper in a quarter. Abnormal stock return and institutional ownership are measured at the quarterly level, and all other control variables are measured at the annual level. The appendix provides the detailed definitions of the variables. The numbers in parentheses denote *p*-values for the test of difference in terms of mean (median) values between the online and print editions. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Summary statistics for news characteristics in the online and print editions

Variable	Online edition: (a)			Print edition: (b)			Test of Difference: (a – b)	
	N	Mean	Median	N	Mean	Median	Mean	Median
Whole article								
Tone	13,676	-1.091	-0.995	12,218	-1.172	-1.039	0.081*** (0.000)	0.044*** (0.000)
Frac neg	13,676	2.139	2.030	12,218	2.177	2.030	-0.038*** (0.007)	0.000 (0.378)
Frac pos	13,676	1.047	0.988	12,218	1.005	0.939	0.042*** (0.000)	0.049*** (0.000)
Title and lead paragraph								
Tone	13,676	-1.355	-1.228	12,218	-1.397	-1.174	0.042 (0.172)	-0.054 (0.646)
Frac neg	13,676	2.530	2.283	12,218	2.464	2.128	0.066*** (0.009)	0.155*** (0.000)
Frac pos	13,676	1.175	0.960	12,218	1.067	0.812	0.108*** (0.000)	0.148*** (0.000)
Coverage	13,676	8.526	3.000	12,218	5.489	2.000	3.037*** (0.000)	1.000*** (0.000)

Panel B: Summary statistics for firm level variables

Variable	N	Mean	Median
Total assets	4,603	65,703	16,336
Sales growth rate	4,565	0.083	0.055
CAPEX / assets	4,589	0.039	0.027
Tobin's q	4,597	2.106	1.668
ROE	4,597	0.011	0.052
Leverage	4,584	0.768	0.643
Abnormal stock return	11,343	0.382	0.295
Institutional ownership	14,608	0.689	0.769

Table 3
Relative Tone of Online News

This table presents the estimates of ordinary least squares (OLS) regressions of the article characteristics on the indicator for online article. The sample includes news articles issued in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Tone* is the number of positive words minus the number of negative words over the total number of words in the article (or title and lead paragraph) as a percentage. *Frac neg* is the number of negative words over the total number of words in the article (or title and lead paragraph) as a percentage. *Frac pos* is the number of positive words over the total number of words in the article (or title and lead paragraph) as a percentage. I partition my sample into an extremely high group and an extremely low group by the value of the news sentiment variables. An article belongs to the extremely high (low) tone group if its *tone* falls into the top (bottom) 20% tone by each year and newspaper. An article belongs to the extremely high (low) frac neg group if the fraction of negative words in the article falls into the top (bottom) 20% *Frac neg* by each year and newspaper. An article belongs to the extremely high (low) frac pos group if the fraction of positive words in the article falls into the top (bottom) 20% *Frac pos* by each year and newspaper. The title and lead paragraph are classified on the basis of Factiva field tags. *Online* is an indicator variable that takes the value of one if the article is published in the online edition and zero otherwise. The numbers in parentheses are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Whole article

Independent variable	Tone			Fraction of negative words			Fraction of positive words		
	Tone	Extremely high tone	Extremely low tone	Frac neg	Extremely high frac neg	Extremely low frac neg	Frac pos	Extremely high frac pos	Extremely low frac pos
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Online (indicator)	0.061*** (3.85)	0.015** (1.98)	-0.038*** (-3.38)	-0.036** (-2.42)	0.030*** (2.66)	-0.004 (-1.09)	0.025*** (6.95)	0.013*** (2.67)	-0.004*** (-2.85)
Log (assets)	-0.056 (-0.92)	-0.042** (-2.44)	0.116 (1.40)	0.053 (0.96)	-0.095 (-1.27)	0.028*** (2.76)	-0.003 (-0.18)	-0.038*** (-2.93)	0.006 (1.56)
Sales growth rate	0.034*** (4.88)	0.006 (0.65)	0.014 (1.43)	-0.041*** (-5.92)	-0.002 (-0.21)	-0.004** (-2.45)	-0.007*** (-2.81)	0.015*** (4.13)	-0.001* (-1.93)
CAPEX / assets	-1.611 (-1.41)	-0.120 (-0.45)	-2.910 (-1.45)	1.323 (1.33)	2.572 (1.46)	0.146 (1.23)	-0.289 (-0.97)	-0.253 (-0.88)	-0.052 (-0.57)
Tobin's <i>q</i>	0.095*** (3.46)	0.005 (0.46)	0.099** (2.10)	-0.084*** (-3.37)	-0.081* (-1.83)	-0.006 (-0.98)	0.012 (1.63)	-0.002 (-0.29)	-0.0002 (-0.11)
ROE	0.006 (1.54)	0.001 (0.15)	-0.0004 (-0.04)	0.002 (0.64)	0.003 (0.37)	0.005*** (3.31)	0.008*** (4.38)	0.003* (1.93)	-0.0001 (-0.42)
Leverage	-0.00003 (-0.12)	-0.0001 (-0.53)	-0.001 (-1.59)	0.0002 (0.90)	0.0004 (0.84)	0.0002*** (2.66)	0.0002*** (3.93)	0.00004 (0.35)	0.0002 (1.18)
Abnormal stock return	0.012*** (5.05)	0.003*** (2.94)	-0.002 (-1.33)	-0.010*** (-4.96)	0.002* (0.90)	-0.001*** (-3.28)	0.002*** (3.51)	0.001* (0.68)	-0.0003* (1.78)
Institutional ownership	-0.398** (-2.05)	0.061** (2.00)	-0.476 (-1.36)	0.386** (2.11)	0.397 (1.26)	-0.007 (-0.49)	-0.012 (-0.43)	-0.018 (-0.48)	-0.0001 (-0.01)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	272,437	52,108	55,688	272,437	55,805	53,131	272,437	54,967	54,418
R-squared	0.081	0.084	0.074	0.082	0.073	0.198	0.044	0.095	0.056

Panel B: Title and lead paragraph

Independent variable	Tone			Fraction of negative words			Fraction of positive words		
	Tone	Extremely high tone	Extremely low tone	Frac neg	Extremely high frac neg	Extremely low frac neg	Frac pos	Extremely high frac pos	Extremely low frac pos
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Online (indicator)	0.039 (1.54)	0.029 (0.82)	-0.237*** (-7.97)	0.006 (0.24)	0.251*** (8.12)	-0.055*** (-4.37)	0.044*** (4.49)	0.076** (2.19)	-0.014*** (-2.87)
Log (assets)	-0.029 (-0.33)	0.060 (1.23)	0.263 (1.63)	0.046 (0.59)	-0.217 (-1.64)	-0.002 (-0.08)	0.017 (0.53)	-0.038 (-0.88)	-0.001 (-0.06)
Sales growth rate	0.030*** (2.79)	0.031 (1.60)	-0.001 (-0.07)	-0.045*** (-4.63)	0.015 (1.03)	0.001 (0.17)	-0.016*** (-3.65)	-0.002 (-0.16)	-0.003** (-2.02)
CAPEX / assets	-3.821** (-2.16)	-1.565* (-1.78)	-4.499** (-2.08)	2.325* (1.81)	3.547* (1.78)	0.015 (0.04)	-0.956* (-1.90)	-1.841* (-1.91)	-0.172 (-0.87)
Tobin's <i>q</i>	0.132*** (3.20)	0.044 (1.64)	0.109 (1.27)	-0.110*** (-2.96)	-0.068 (-0.85)	0.010 (0.83)	0.022* (1.84)	0.021 (0.71)	0.001 (0.11)
ROE	0.009 (1.64)	-0.011 (-1.34)	0.006 (0.41)	0.003 (0.72)	0.004 (0.27)	0.009*** (2.82)	0.012*** (7.50)	0.009 (0.68)	-0.0004 (-0.47)
Leverage	0.0004 (0.79)	-0.001 (-1.27)	0.002*** (3.30)	-0.0004 (-0.75)	-0.003*** (-2.70)	0.001** (2.27)	0.00001 (0.16)	-0.001*** (3.68)	-0.0002 (-0.33)
Abnormal stock return	0.016*** (4.11)	0.0002 (0.06)	-0.002 (-0.21)	-0.013*** (-4.02)	0.001 (0.29)	-0.001 (-0.48)	0.003*** (2.79)	0.003 (1.05)	-0.001 (-1.61)
Institutional ownership	-0.475* (-1.87)	0.086 (0.75)	-0.536 (-1.10)	0.479** (2.01)	0.423 (0.92)	-0.017 (-0.47)	0.004 (0.08)	0.076 (1.05)	0.002 (0.10)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	272,437	52,108	55,688	272,437	55,805	53,131	272,437	54,967	54,418
R-squared	0.041	0.043	0.034	0.044	0.032	0.043	0.027	0.070	0.018

Table 4
Tone Volatility and Investor Attention

This table presents the estimates of ordinary least squares (OLS) regressions of the article characteristics on the indicator for online article. The sample includes news articles issued in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Tone volatility* is the standard deviation in the *tone* of articles issued by a newspaper in a quarter. *Frac neg volatility* is the standard deviation in the *frac neg* of articles issued by the newspaper in a quarter. *Frac pos volatility* is the standard deviation in the *frac pos* of articles issued by the newspaper in a quarter. *Online* is an indicator variable that takes the value one if the article is published in the online edition and zero otherwise. The sample is divided into quartiles based on the values of share turnover, extreme abnormal return, and media coverage. The numbers in parentheses are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Relative tone volatility of online news

Independent variable	Tone volatility (1)	Frac neg volatility (2)	Frac pos volatility (3)
Online (indicator)	0.048*** (4.60)	0.044*** (5.00)	0.031*** (7.04)
Control variables as in Table 3	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes
Observations	21,559	21,559	21,559
<i>R</i> -squared	0.108	0.111	0.109

Panel B: Tone volatility and investor attention

Independent variable	Share Turnover		Extreme Abnormal Return		Media Coverage	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)
Panel B1: Tone volatility						
Online (indicator)	0.077*** (4.12)	0.025 (1.34)	0.089*** (4.48)	0.036** (2.00)	0.047*** (3.80)	-0.012 (-0.37)
Test of difference between high and low in online: (p-value)	0.052** (0.048)		0.053** (0.049)		0.059* (0.089)	
R-squared	0.178	0.156	0.181	0.189	0.251	0.118
Panel B2: Frac neg volatility						
Online (indicator)	0.064*** (3.95)	0.021 (1.33)	0.068*** (3.89)	0.042*** (2.90)	0.045*** (4.13)	-0.004 (-0.14)
Test of difference between high and low in online: (p-value)	0.043* (0.054)		0.026 (0.256)		0.049 (0.104)	
R-squared	0.176	0.161	0.187	0.203	0.273	0.115
Panel B3: Frac pos volatility						
Online (indicator)	0.054*** (6.59)	0.023** (2.51)	0.045*** (5.99)	0.020** (2.31)	0.026*** (5.51)	0.015 (1.06)
Test of difference between high and low in online: (p-value)	0.031** (0.010)		0.025** (0.030)		0.011 (0.469)	
R-squared	0.180	0.141	0.217	0.170	0.269	0.131
Control variables as in Table 3	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,395	5,386	5,378	5,383	5,612	5,046

Table 5
Market Response to Online and Print News

This table presents the estimates of ordinary least squares (OLS) regressions of two-day cumulative abnormal returns (CAR [0,1]) around article issuance date on online article tone and print article tone. The sample includes news articles issued in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Online tone* is the average daily tone of the articles (or titles and lead paragraphs) issued in the online editions. *Online frac neg* is the average daily fraction of negative words of the articles (or titles and lead paragraphs) in the online editions. *Online frac pos* is the average daily fraction of positive words of the articles (or titles and lead paragraphs) in the online editions. *Print tone* is the average daily tone of the articles (or titles and lead paragraphs) issued in the print editions. *Print frac neg* is the average daily fraction of negative words in the articles (or titles and lead paragraphs) in the print editions. *Print frac pos* is the average daily fraction of positive words in the articles (or titles and lead paragraphs) in the print editions. The title and lead paragraph are classified on the basis of Factiva field tags. The numbers in parentheses are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Independent variable	CAR [0,1]					
	Whole Article			Title and Lead Paragraph		
	(1)	(2)	(3)	(4)	(5)	(6)
Online tone	0.058*** (8.88)			0.044*** (10.29)		
Print tone	0.008 (1.12)			0.006 (1.36)		
Online frac neg		-0.045*** (-7.22)			-0.038*** (-8.81)	
Print frac neg		-0.015** (-2.14)			-0.011** (-2.23)	
Online frac pos			0.083*** (5.99)			0.065*** (7.45)
Print frac pos			0.009 (0.54)			-0.002 (-0.32)
Control variables as in Table 3	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Test of difference: Online - Print (p-value)	0.050*** (0.000)	-0.030*** (0.000)	0.074*** (0.000)	0.038*** (0.000)	-0.027*** (0.000)	0.067*** (0.000)
Observations	106,357	106,357	106,357	106,357	106,357	106,357
R-squared	0.011	0.010	0.010	0.011	0.011	0.010

Table 6
Relative Coverage of Online News

This table presents the estimates of ordinary least squares (OLS) regressions of *Coverage* on the indicator for online article. The sample includes news articles issued in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Coverage* is the number of articles issued by the newspaper in a quarter. *Online* is an indicator variable that takes the value of one if the article is published in the online edition and zero otherwise. The numbers in parentheses are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Independent variable	Coverage (1)
Online (indicator)	3.677*** (8.58)
Log (assets)	5.355*** (2.96)
Sales growth rate	2.085* (1.89)
CAPEX / assets	-19.060 (-0.96)
Tobin's <i>q</i>	2.105*** (4.31)
ROE	-2.457*** (-4.79)
Leverage	0.0002 (0.05)
Abnormal stock return	-0.059 (-1.60)
Institutional ownership	-4.518 (-1.35)
Firm fixed effects	Yes
Quarter fixed effects	Yes
Newspaper fixed effects	Yes
Observations	32,193
<i>R</i> -squared	0.457

Table 7
Same Title Articles by the Same Journalist(s)

This table presents the summary statistics (Panel A) and estimates of ordinary least squares (OLS) regressions of the article characteristics on the indicator for online article (Panel B). The sample includes 23,762 same-title articles by the same journalist(s) published in the online and print editions of WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Tone* is the number of positive words minus the number of negative words over the total number of words as a percentage. *Frac neg* is the number of negative words over the total number of words as a percentage. *Frac pos* is the number of positive words over the total number of words as a percentage. The title and lead paragraph are classified on the basis of Factiva field tags. *Online* is an indicator variable that takes the value of one if the article is published in the online edition and zero otherwise. The numbers in parentheses (Panel A) denote *p*-values for the test of difference in terms of mean (median) values between the online and print editions. The numbers in parentheses (Panel B) are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Summary statistics for news characteristics in the online and print edition

Variable	Online edition (N = 11,881): (a)		Print edition (N = 11,881): (b)		Test of Difference: (a – b)	
	Mean	Median	Mean	Median	Mean	Median
Title						
Tone	-0.026	0.000	-0.026	0.000	0.000 (1.000)	0.000 (1.000)
Frac neg	3.707	0.000	3.707	0.000	0.000 (1.000)	0.000 (1.000)
Frac pos	1.121	0.000	1.121	0.000	0.000 (1.000)	0.000 (1.000)
Lead paragraph						
Tone	-1.588	-0.971	-1.485	0.000	-0.103** (0.015)	-0.971** (0.044)
Frac neg	2.466	1.667	2.345	1.539	0.121*** (0.001)	0.128*** (0.000)
Frac pos	0.878	0.000	0.860	0.000	0.018 (0.330)	0.000* (0.076)
Text following lead paragraph						
Tone	-1.203	-0.872	-1.190	-0.867	-0.013 (0.573)	-0.005 (0.573)
Frac neg	2.161	1.795	2.140	1.773	0.021 (0.301)	0.022 (0.215)
Frac pos	0.957	0.865	0.950	0.855	0.007 (0.378)	0.010 (0.125)

Panel B: Relative tone of online news

Independent variable	Tone					
	Whole article			Title and lead paragraph		
	Tone	Frac neg	Frac pos	Tone	Frac neg	Frac pos
	(1)	(2)	(3)	(4)	(5)	(6)
Online (indicator)	-0.007** (-2.08)	0.014*** (4.66)	0.007*** (3.95)	-0.065*** (-4.63)	0.084*** (7.52)	0.018** (2.40)
Log (assets)	-0.123 (-1.21)	0.072 (0.84)	-0.051* (-1.81)	-0.101 (-0.58)	0.058 (0.40)	-0.042 (-0.81)
Sales growth rate	0.044*** (2.76)	-0.022 (-1.18)	0.022** (2.07)	0.027 (0.89)	0.014 (0.43)	0.040*** (3.15)
CAPEX / assets	-2.696 (-1.52)	2.078 (1.27)	-0.617 (-0.92)	-6.090* (-1.81)	2.939 (1.05)	-3.151*** (-2.62)
Tobin's <i>q</i>	-0.059 (-1.00)	0.033 (0.63)	-0.026 (-1.18)	-0.181 (-1.57)	0.110 (1.05)	-0.070** (-2.06)
ROE	0.017** (2.11)	-0.017** (-2.03)	0.001 (0.08)	0.009 (0.40)	0.002 (0.08)	0.010*** (2.70)
Leverage	-0.001 (-1.43)	0.001*** (2.91)	0.000 (0.47)	-0.000 (-0.44)	0.000 (0.04)	-0.000 (-0.36)
Abnormal stock return	0.011** (2.42)	-0.010*** (-2.81)	0.001 (0.47)	0.008 (0.94)	-0.009 (-1.10)	-0.001 (-0.20)
Institutional ownership	-0.269 (-0.84)	0.271 (1.01)	0.002 (0.03)	0.066 (0.14)	0.026 (0.06)	0.092 (0.90)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,740	17,740	17,740	17,740	17,740	17,740
R-squared	0.141	0.147	0.083	0.087	0.094	0.060

Table 8
Online News Trading Profits

This table presents the daily risk-adjusted returns from an equally weighted or value-weighted long-short portfolio. Panel A (B) presents the daily risk-adjusted returns if the investment is made without (with) waiting after portfolio construction. The sample includes 17,008 same-title articles by the same journalist(s) published on the same date in online and print editions of WSJ, NYT, USA Today, and WSP from 2007 to 2018. Each day, stocks are sorted on the basis of abnormal online tone. Abnormal online tone is calculated as the difference between the online tone and print tone of the title and lead paragraph. I define high (low) abnormal online tone by the value of the sample median. I long (short) stocks with low (high) abnormal online tone and hold both long and short positions for one day after the investment. The portfolio is rebalanced daily. I examine three different factor models using the market model, the Fama and French (1993) three-factor model, and Carhart's (1997) four-factor model. The title and lead paragraph are classified on the basis of Factiva field tags. The appendix provides detailed definitions of the variables. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Investment without waiting after portfolio construction

Independent variable	Equally weighted portfolios			Value-weighted portfolios		
	(1)	(2)	(3)	(4)	(5)	(6)
RM-RF	-0.0212 (-1.52)	-0.0303* (-1.86)	-0.0292* (-1.80)	-0.0423*** (-2.83)	-0.0514*** (-2.97)	-0.0514*** (-2.90)
SMB		0.0491 (1.59)	0.0534* (1.73)		0.0593* (1.80)	0.0644* (1.96)
HML		-0.0059 (-0.21)	0.0322 (1.04)		-0.0171 (-0.57)	0.0278 (0.84)
UMD			0.0561*** (2.82)			0.0663*** (3.12)
Alpha	0.0005*** (3.24)	0.0005*** (3.26)	0.0005*** (3.50)	0.0004*** (2.59)	0.0004*** (2.61)	0.0004*** (2.65)
Observations	1,688	1,688	1,688	1,688	1,688	1,688
R-squared	0.0014	0.0029	0.0076	0.0047	0.0069	0.0126

Panel B: Excess return for the investment with waiting after portfolio construction

Waiting days after portfolio construction	Equally weighted portfolios			Value-weighted portfolios		
	(1)	(2)	(3)	(4)	(5)	(6)
1	0.0003** (2.52)	0.0003** (2.54)	0.0003** (3.54)	0.0002* (1.79)	0.0002* (1.80)	0.0002* (1.80)
2	0.0003** (2.05)	0.0003** (2.00)	0.0003** (1.99)	0.0002 (1.39)	0.0002 (1.32)	0.0002 (1.32)
3	0.0001 (0.77)	0.0001 (0.76)	0.0001 (0.76)	0.0001 (0.72)	0.0001 (0.72)	0.0001 (0.73)
Control variables as in Panel A	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,688	1,688	1,688	1,688	1,688	1,688

Table 9
Post Earnings Announcements Tone

This table presents the estimates of ordinary least squares (OLS) regressions of article characteristics on the indicator for online article. The sample includes news articles issued within one day of earnings announcements in online and print editions by WSJ, NYT, USA Today, and WSP from 2007 to 2018. *Tone* is the number of positive words minus the number of negative words over the total number of words in the title and lead paragraph as a percentage. *Frac neg* is the number of negative words over the total number of words in the title and lead paragraph as a percentage. *Frac pos* is the number of positive words over the total number of words in the title and lead paragraph as a percentage. The title and lead paragraph are classified on the basis of Factiva field tags. Positive announcements are when the value of actual earnings minus the mean value of the analyst forecasts is greater or equal to 5 cents. Neutral announcements are when the value of actual earnings minus the mean value of the analyst forecasts is greater or equal to 0 cents and less than 5 cents. Missed announcements are when the value of actual earnings minus the mean value of the analyst forecasts is less than 0 cents. *Online* is an indicator variable that takes the value of one if the article is published in the online edition and zero otherwise. The numbers in parentheses are *t*-statistics based on robust standard errors adjusted for clustering by firm. The appendix provides detailed definitions of the variables. Constant is included in the regression but not reported for brevity. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Independent variable	Title and lead paragraph								
	Positive Announcements			Neutral Announcements			Missed Announcements		
	Tone (1)	Frac neg (2)	Frac pos (3)	Tone (4)	Frac neg (5)	Frac pos (6)	Tone (7)	Frac neg (8)	Frac pos (9)
Online (indicator)	0.117** (1.99)	0.078 (1.58)	0.194*** (5.79)	0.039 (0.57)	0.134*** (2.78)	0.173*** (3.79)	-0.174** (-2.06)	0.287*** (3.53)	0.114*** (2.71)
Forecast dispersion	-0.934 (-0.60)	1.073 (0.77)	0.139 (0.24)	-7.389* (-1.92)	4.979* (1.90)	-2.410 (-1.22)	0.343 (0.31)	-0.222 (-0.24)	0.121 (0.26)
Control variables as in Table 3	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Newspaper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,964	9,964	9,964	8,402	8,402	8,402	5,172	5,172	5,172
R-squared	0.106	0.117	0.100	0.117	0.114	0.115	0.156	0.159	0.141

Appendix

Variable Definitions and Sources

CCM denotes Compustat/CRSP Merged database.

Variable	Definition	Source
Tone	The number of positive words minus the number of negative words over the total number of words in the article (or title and lead paragraph) as a percentage.	Factiva
Frac neg	The number of negative words over the total number of words in the article (or title and lead paragraph) as a percentage.	Factiva
Frac pos	The number of positive words over the total number of words in the article (or title and lead paragraph) as a percentage.	Factiva
Tone volatility	The standard deviation in the <i>tone</i> of articles issued by the newspaper in a quarter.	Factiva
Frac neg volatility	The standard deviation in the <i>frac neg</i> of articles issued by the newspaper in a quarter.	Factiva
Frac pos volatility	The standard deviation in the <i>frac pos</i> of articles issued by the newspaper in a quarter.	Factiva
Coverage	The number of articles issued by the newspaper in a quarter.	Factiva
Online (indicator)	An indicator variable that takes the value of one if the article is published in the online edition and zero otherwise.	Factiva
Online tone	The average daily tone of the articles (or titles and lead paragraphs) issued in the online edition.	Factiva
Online frac neg	The average daily fraction of negative words in the articles (or titles and lead paragraphs) in the online edition.	Factiva
Online frac Pos	The average daily fraction of positive words in the articles (or titles and lead paragraphs) in the online edition.	Factiva
Print tone	The average daily tone of the articles (or titles and lead paragraphs) issued in the print edition.	Factiva
Print frac neg	The average daily fraction of negative words in the articles (or titles and lead paragraphs) in the print edition.	Factiva
Print frac Pos	The average daily fraction of positive words in the articles (or titles and lead paragraphs) in the print edition.	Factiva
Abnormal online tone	The difference between the online tone and print tone of the title and lead paragraph.	Factiva
Log (assets)	Natural logarithm of total assets.	CCM
Sales growth rate	The growth rate of sales between the current and previous fiscal year.	CCM
CAPEX / assets	Capital expenditure over total assets.	CCM

Tobin's q	The market value of the firm over total assets.	CCM
ROE	Net income over shareholder equity.	CCM
Leverage	Total debt over total assets.	CCM
Abnormal stock return	Value weighted abnormal stock return in a quarter.	CRSP
Share turnover	The average value of trading volume over the number of shares outstanding.	CRSP
Institutional ownership	The proportion of shares held by institutions.	Thomson-Reuters Institutional Holdings (13F)
RM-RF	The excess return on the market.	Kenneth French Data Library
SMB	The average return on the portfolios with small stocks minus the average return on the portfolios with big stocks (Fama and French (1993)).	Kenneth French Data Library
HML	The average return on the portfolios with high book-to-market ratio stocks minus the average return on the portfolios with low book-to-market ratio stocks.	Kenneth French Data Library
UMD	The average return on the portfolios with high past 12-month return stocks minus the average return on the portfolios with low past 12-month return stocks.	Kenneth French Data Library
Forecast error	The value of actual earnings minus the mean value of analyst forecasts.	I/B/E/S
Forecast dispersion	The standard deviation of analyst forecasts.	I/B/E/S